

An Information-Theoretic Framework for Emergent Spacetime Dynamics, Horizon Thermodynamics, and Testable Cosmological Signatures

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Abstract

A central challenge in theoretical physics is to understand whether quantum-information concepts, thermodynamic constraints, and classical gravitational dynamics can be placed within a common mathematical architecture without conflating their distinct domains of validity. We present a four-bridge theoretical framework that separates established results from explicit modeling assumptions. First, a global quantum state subject to a Wheeler–DeWitt-type constraint is partitioned, as a structural assumption, into observer and environment subsystems; standard partial tracing then defines the observer reduced state and its von Neumann entropy. Second, thermodynamic entropy is distinguished from von Neumann entropy expressed in bits, and Landauer’s bound is applied specifically to logically irreversible erasure or reset operations. Third, we introduce a dimensionally closed phenomenological dynamical system with separate model-information and irreversible-erasure rates. Fourth, we review the Bekenstein–Hawking area–entropy relation and Jacobson’s local horizon-thermodynamic recovery of the Einstein equation, while leaving the microscopic identification between observer information variables and horizon degrees of freedom explicitly open. Motivated by logarithmic corrections to black-hole entropy in quantum-gravity literature, we finally introduce a log-periodic modulation of a primordial gravitational-wave spectrum as a candidate phenomenological observational ansatz. The mapping from the internal dynamical variables to the modulation parameters is not derived here and remains an open problem.

Keywords: Quantum Information, Horizon Thermodynamics, Emergent Gravity, Holographic Principle, Landauer’s Principle, Primordial Gravitational Waves.

1 Introduction

Information-theoretic and thermodynamic approaches to gravity motivate the question of whether relationships among quantum states, entropy, causal horizons, and spacetime geometry can be organized within a common framework [1, 7, 8]. The purpose of this paper is not to identify these concepts as physically equivalent. Instead, it constructs an explicit architecture in which established physical results, applications under stated assumptions, and new phenomenological hypotheses are kept separate.

The framework contains four conceptual bridges. The *quantum bridge* begins with a Wheeler–DeWitt-type constraint and introduces an observer–environment tensor factorization as a modeling assumption. The *thermodynamic bridge* distinguishes physical von Neumann entropy from its dimensionless representation in bits and restricts Landauer’s bound to logically irreversible erasure or reset. The *dynamical bridge* introduces a finite-resource phenomenological ODE system with explicit units. The *cosmological bridge* places the model alongside horizon

area–entropy relations and Jacobson’s local horizon thermodynamics, but does not identify the observer entropy with horizon entropy or local Landauer dissipation with horizon heat flux.

A final section introduces a log-periodic gravitational-wave spectral modulation as a candidate observational ansatz motivated by logarithmic corrections to horizon entropy. This spectral form is not claimed to follow uniquely from the preceding dynamical equations.

2 The Quantum Bridge: Subsystem Partitioning and Reduced State

In canonical quantum gravity, the Wheeler–DeWitt constraint is conventionally represented schematically as

$$\hat{H}|\Psi\rangle = 0, \quad (1)$$

with important interpretational and technical qualifications in the full theory [2]. In the present framework, we additionally *assume* that the relevant Hilbert space admits an observer–environment tensor decomposition,

$$\mathcal{H}_{\text{total}} = \mathcal{H}_{\text{obs}} \otimes \mathcal{H}_{\text{env}}. \quad (2)$$

This factorization is a structural modeling assumption; it is not derived from the Wheeler–DeWitt constraint.

Given the factorization, a pure global state can be expanded as

$$|\Psi\rangle = \sum_{i,j} c_{ij} |\phi_i\rangle_{\text{obs}} \otimes |\chi_j\rangle_{\text{env}}, \quad (3)$$

where the two sets of kets form orthonormal bases for their respective factors. Standard quantum mechanics then defines the observer reduced density operator by the partial trace,

$$\rho_{\text{obs}} = \text{Tr}_{\text{env}}(|\Psi\rangle\langle\Psi|) = \sum_j \langle\chi_j|\Psi\rangle\langle\Psi|\chi_j\rangle. \quad (4)$$

The corresponding physical von Neumann entropy is

$$S_{\text{vN}} = -k_B \text{Tr}(\rho_{\text{obs}} \ln \rho_{\text{obs}}), \quad [S_{\text{vN}}] = \text{J K}^{-1}. \quad (5)$$

For a globally pure bipartite state, the reduced entropy quantifies entanglement across the chosen bipartition; for more general mixed global states, reduced entropy need not represent entanglement alone.

For the dynamical model below, we define the same von Neumann entropy expressed in bits,

$$H_{\text{obs}} \equiv \frac{S_{\text{vN}}}{k_B \ln 2} = -\text{Tr}(\rho_{\text{obs}} \log_2 \rho_{\text{obs}}). \quad (6)$$

Here H_{obs} is dimensionless in physical dimensions, with its numerical value expressed in bits. Within the proposed framework it is used as an information-theoretic state variable associated with observer–environment correlations. Identifying this variable with macroscopic organizational or self-modeling capacity is an explicit modeling hypothesis rather than a consequence of the Wheeler–DeWitt constraint or the partial-trace construction.

3 The Thermodynamic Bridge: Landauer Bound and Rate Separation

Landauer's principle associates logical irreversibility with a thermodynamic cost [3]. For an ideal one-bit erasure or reset at temperature T , the limiting energy scale is

$$E_{\min} = k_B T \ln 2. \quad (7)$$

Accordingly, if $\dot{I}_{\text{erase}}$ denotes the rate of logically irreversible bit erasures or resets, the corresponding dissipative power obeys the bound

$$\mathcal{P}_{\text{diss}} \geq k_B T \ln 2 \dot{I}_{\text{erase}}. \quad (8)$$

Equality represents an ideal limiting case and is not assumed as a generic dynamical law.

We distinguish $\dot{I}_{\text{erase}}$ from a separate phenomenological internal information rate $\dot{I}_{\text{model}}$. No universal identity between these rates is assumed. Any relation between them is model dependent and constitutes an additional hypothesis. Landauer's bound therefore constrains the irreversible-erasure channel; by itself it does not prove a universal ceiling on generic information processing, organizational complexity, or self-modeling.

4 Phenomenological Dynamical Model and Dimensional Closure

We now introduce a phenomenological finite-resource dynamical ansatz. Let γ denote a reconstruction or relaxation-rate variable, $\gamma_{\max}$ its saturation scale, $\mathcal{E}_{\text{avail}}$ an available energy scale, and $\mathcal{P}_{\text{input}}$ an external input power. Let $H_{\max}$ denote an informational saturation parameter in bits. The proposed system is

$$\frac{d\gamma}{dt} = a \dot{I}_{\text{model}} \mathcal{E}_{\text{avail}} \left(1 - \frac{\gamma}{\gamma_{\max}} \right) - b\gamma, \quad (9)$$

$$\frac{dH_{\text{obs}}}{dt} = c\gamma (H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}}, \quad (10)$$

$$\frac{d\dot{I}_{\text{model}}}{dt} = e \frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{input}} - g\dot{I}_{\text{model}}. \quad (11)$$

These equations are not derived from non-equilibrium statistical mechanics; they define the phenomenological dynamical sector of the proposed architecture.

The dimensions are assigned as

$$\begin{aligned} [H_{\text{obs}}] &= [H_{\max}] = 1 \text{ (bits)}, & [\dot{I}_{\text{model}}] &= [\dot{I}_{\text{erase}}] = \text{s}^{-1}, \\ [\gamma] &= [\gamma_{\max}] = \text{s}^{-1}, & [\mathcal{E}_{\text{avail}}] &= \text{J}, \\ [\mathcal{P}_{\text{input}}] &= [\mathcal{P}_{\text{diss}}] = \text{J s}^{-1}. \end{aligned} \quad (12)$$

Dimensional homogeneity then requires

$$[a] = [f] = \text{J}^{-1} \text{s}^{-1}, \quad [b] = [e] = [g] = \text{s}^{-1}, \quad [c] = 1. \quad (13)$$

Thus a and f have dimensions of inverse action. Within this finite-resource ansatz, irreversible-processing costs can contribute to saturation when a specified relationship between $\dot{I}_{\text{model}}$ and $\dot{I}_{\text{erase}}$ is supplied; Landauer's bound alone does not determine that relationship.

5 The Cosmological Bridge: Horizon Thermodynamics and an Open Domain Extension

5.1 Horizon area–entropy relation and informational capacity

Black-hole thermodynamics associates horizon area with entropy [4, 5]. Written in physical entropy units, the Bekenstein–Hawking expression is

$$S_{\text{BH}} = \frac{k_B A}{4\ell_P^2}, \quad \ell_P = \sqrt{\frac{\hbar G}{c^3}}. \quad (14)$$

Expressed in bits, the corresponding quantity is

$$H_{\text{BH}} = \frac{S_{\text{BH}}}{k_B \ln 2} = \frac{A}{4\ell_P^2 \ln 2}. \quad (15)$$

Covariant and holographic entropy bounds provide broader gravitational context for area-based information limits [6, 7].

For the present phenomenological model, we introduce the boundary condition

$$H_{\text{obs}} \leq H_{\text{max}}, \quad H_{\text{max}} \equiv \frac{A}{4\ell_P^2 \ln 2}, \quad (16)$$

for a specified causal-horizon area A . Applying the horizon area–entropy scale to this observer subsystem is a structural domain extension of the model, not a derivation that identifies H_{obs} with microscopic horizon entropy. The microphysical mapping between observer degrees of freedom and gravitational horizon degrees of freedom remains open.

5.2 Jacobson’s local horizon thermodynamics

Independently of the observer ODE model above, Jacobson showed that the Einstein equation can be recovered as an equation of state by imposing the Clausius relation on local Rindler causal horizons, together with entropy proportional to horizon area and the Unruh temperature [8]. Schematically, the heat flux through a local causal horizon is written

$$\delta Q = \int T_{\mu\nu} \xi^\mu d\Sigma^\nu, \quad (17)$$

and the thermodynamic relation

$$\delta Q = T dS \quad (18)$$

is combined with the horizon area response governed locally by the Raychaudhuri equation. Under Jacobson’s stated assumptions and using stress-energy conservation together with the contracted Bianchi identity, the result takes the Einstein form

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (19)$$

This established horizon-thermodynamic result does not by itself identify the model’s H_{obs} with horizon entropy, nor $\mathcal{P}_{\text{diss}}$ with the horizon heat flux $\delta Q/dt$. Those possible connections remain open theoretical bridges.

6 Candidate Cosmological Signature: A Phenomenological Ansatz

Logarithmic corrections to the leading Bekenstein–Hawking entropy occur in quantum-gravity calculations; for example, Kaul and Majumdar obtain a logarithmic contribution together with subleading terms in a quantum-geometry treatment of black-hole entropy [9]. Motivated by that class of corrections, consider the schematic physical-entropy form

$$S_{\text{corr}} = \frac{k_B A}{4\ell_P^2} - \alpha k_B \ln\left(\frac{A}{4\ell_P^2}\right) + \mathcal{O}(k_B), \quad (20)$$

where α is treated here as a phenomenological dimensionless coefficient rather than a universal constant.

Normalizing to bits gives

$$H_{\text{corr}} \equiv \frac{S_{\text{corr}}}{k_B \ln 2} = \frac{A}{4\ell_P^2 \ln 2} - \frac{\alpha}{\ln 2} \ln\left(\frac{A}{4\ell_P^2}\right) + \mathcal{O}(1). \quad (21)$$

The literature motivation for a logarithmic entropy correction does not uniquely determine a primordial gravitational-wave spectrum in the present architecture.

As a candidate observational parameterization, we therefore introduce the phenomenological ansatz

$$P_{\text{GW}}(k) = P_0(k) [1 + \epsilon \cos(\beta \ln(k/k_0) + \phi_0)], \quad (22)$$

where $P_0(k)$ is a baseline spectrum and $(\epsilon, \beta, \phi_0, k_0)$ are phenomenological parameters. For numerical illustrations one may choose, for example, $\epsilon = 10^{-4}$ as a benchmark amplitude; this value is not derived by the framework and is not asserted as a measured or theoretically fixed constant. No relation such as $\beta = 2\pi\alpha$ is assumed without an additional derivation.

Equation (20) is therefore a falsifiable spectral template, not a first-principles consequence of the ODE system. A complete predictive theory would need to derive a map from the entropy correction and cosmological background dynamics to the tensor perturbation equations and thereby determine or constrain $(\epsilon, \beta, \phi_0)$.

7 Discussion and Limitations

The architecture deliberately leaves several theoretical boundaries open. First, the observer–environment tensor factorization is an assumed subsystem structure rather than a consequence of the Wheeler–DeWitt constraint. Second, the interpretation of H_{obs} as organizational or self-modeling capacity is phenomenological. Third, Landauer’s principle constrains logically irreversible erasure; a relation between $\dot{I}_{\text{model}}$ and $\dot{I}_{\text{erase}}$ must be separately specified. Fourth, the ODE system is a dimensionally closed phenomenological ansatz, not a derivation from microscopic non-equilibrium dynamics. Fifth, applying a horizon-area capacity scale to the observer subsystem is a domain extension whose microphysical justification remains open. Sixth, Jacobson’s local horizon construction is retained as an established, logically separate result rather than being used to claim that local computational dissipation generates spacetime curvature. Finally, the log-periodic gravitational-wave spectrum is introduced as a candidate observational template; the cosmological transfer mechanism from logarithmic entropy corrections to that spectrum has not been derived here.

These boundaries make clear what observations or future derivations could change the status of the framework. Deriving the ODEs from a microscopic model, constructing an explicit observer-to-horizon map, and obtaining the spectral parameters from a specified cosmological action or perturbation theory would convert presently phenomenological links into stronger theoretical results. Conversely, failure of such derivations or observational constraints excluding relevant parameter regions would restrict the proposed architecture.

8 Conclusion

This paper constructs a defined information-theoretic and thermodynamic architecture while separating established physics from explicit assumptions and phenomenological proposals. Standard quantum reduction supplies ρ_{obs} once an observer–environment factorization is assumed; von Neumann entropy is kept distinct from its bit-normalized representation; Landauer’s principle is restricted to logically irreversible erasure; and the proposed dynamical equations are dimensionally closed with explicit coefficient units. Horizon thermodynamics and Jacobson’s recovery of the Einstein equation provide established gravitational context, while the identification of observer information with horizon degrees of freedom remains open. Logarithmic horizon-entropy corrections motivate, but do not derive, the proposed log-periodic gravitational-wave spectral template. The framework therefore defines a set of explicit theoretical bridges and open problems rather than claiming a completed microscopic theory of emergent spacetime.

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